

HHUflauschig on Candy Speech Detection: Hybrid Approaches for Binary Classification and Span Typing

Wiebke Petersen, Lara Eulenpesch
Heinrich Heine Universität
Düsseldorf, Germany
{wiebke.petersen, lara.eulenpesch}@hhu.de

GermEval 2025 Shared Task on Candy Speech Detection,
Hildesheim

Subtask 1: Task definition

Task:

- **Objective:** Identify whether a German YouTube comment contains *Flausch* (affectionate, positive language).
- **Task Type:** Binary classification.
- **Challenges:** Subjective definitions, informal language, spelling variation.

Subtask 1: System Overview

Data preparation:

- added spelling corrected comments
- added translations
- create held-out 10% split of training data for evaluation

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Approach:

- Hybrid architecture combining:
 - Linguistically motivated features
 - Fine-tuned transformer models
- Final prediction made by meta-classifier (logistic regression).

Fine-tuned LLMs

Results on held-out evaluation data

Model	Input	F1
gbert-large	original	0.906
gbert-large	spelling corrected	0.896
bert-base-german	original	0.885
bert-base-german	spelling corrected	0.880
roberta-large	translated	0.875

Results: original input > corrected text
large models > base models
German models > English model on translations

Features

- **Softmax scores of fine-tuned LLMs:** for original, spelling corrected and translated comments
- **Sentiment Polarity:** via TextBlob and TextBlobDE
- **Ekman's Emotions Scores:** via English translations and available fine-tuned RoBERTa model (*anger, disgust, fear, happiness, sadness, surprise*)
- **Positive Lexicon features:**
 - Lists of positive words, tokens, emojis, emoticons (via ChatGPT-4o)
 - Tokens filtered out by frequency in non-Flausch comments
 - Absolute count and ratio features
- **Surface Features:**
 - Number of words with consecutive capital letters (2+)
 - Number of repeated characters (3+)

Results of meta-classifiers (logistic regression)

Results on held-out evaluation data

Features	F1	Rec.	Prec.
all non-BERT features	0.694	0.785	0.621
all BERT features	0.926	0.944	0.908
all features	0.932	0.936	0.927
winning configuration	0.938	0.929	0.947
gbert-large on orig.	0.906	0.881	0.932

winning configuration = gbert-large orig. comment + all sentiments (Ekman + polarity) + positive word count + positive token count + positive token ratio

Results on competition test data

winning configuration	0.887	0.900	0.875
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BA-thesis of Eulenpesch (with improved features):

Results: all > BERT > positive counts > sentiments

Subtask 2: Task Definition

- **Goal:** Identify text spans expressing *Flausch* and assign one of 10 *Flausch-types*.
- **Evaluation:**
 - **Strict F1:** Correct span boundaries *and* correct type
 - **Span F1:** Span only
 - **Type F1:** Type only
- **Challenge:** Both accurate segmentation and subtle type classification

Subtask 2: System Overview

- We explored two paradigms:
 1. **End-to-End**: single model for joint span+type prediction (fine-tuned gbert-large)
 2. **Two-Step Pipeline**:
 - Step 1: span segmentation (rule-based or with LLM)
 - Step 2: type classification (with LLM)

Subtask 2: System Overview

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 1. **End-to-End:** single model for joint span+type prediction (fine-tuned gbert-large)
 2. **Two-Step Pipeline:**
 - Step 1: span segmentation (rule-based or with LLM)
 - Step 2: type classification (with LLM)
- **Best System:** Two-step pipeline based on gbert-large for segmentation and classification

Two-Step Pipelines: Span Segmentation

LLM Approach: Token-level BIO tagging with BERT

Rule-Based Approach: : We apply heuristics over SpaCy dependency trees:

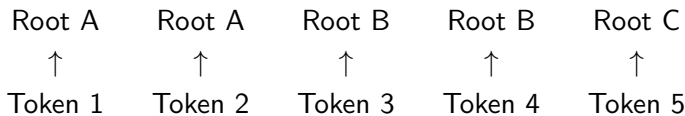
Token 1 Token 2 Token 3 Token 4 Token 5

Two-Step Pipelines: Span Segmentation

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- for each token, we traverse upward until reaching a root, which is either the syntactic root, *reported speech* (rs), *coordinating conjunction* (cd), or *junctor* (ju).

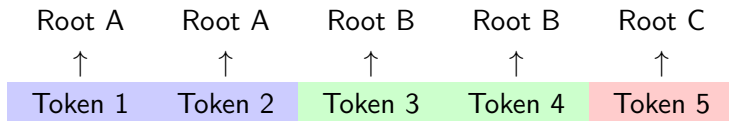


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- Consecutive tokens sharing the same root form a span.



Two-Step Pipelines: Span classification

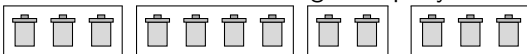
gbert-large fine-tuned to classify spans into

- 10 Flausch types (trained on typed flausch spans) or
- 10 Flausch types + not-Flausch class (trained on typed flausch spans + non-Flausch spans)

Two-Step Pipelines: Span classification

gbert-large fine-tuned to classify spans into

- 10 Flausch types (trained on typed flausch spans) or
- 10 Flausch types + not-Flausch class (trained on typed flausch spans + non-Flausch spans)
- non-Flausch spans are generated from
 - non Flausch comments using our SpaCy heuristics



- Flausch spans by splitting at Flausch spans



Results

Results on held-out evaluation data

System	Strict	Span	Type
gbert-end-to-end	0.647	0.682	0.792
gbert 2-step	0.728	0.769	0.833
gbert 2-step + not-flausch	0.693	0.769	0.785
spacy 2-step	0.370	0.389	0.733

Results on competition test data

gbert 2-step	0.615	0.668	0.766
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- additional not-flausch label for rejecting non-Flausch spans → no improvement
- rule-based approach does not identify correct spans

improved rule-based approach in BA-thesis (still weaker than gbert 2-step)

Some limitations

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- **Data Split:** Held-out evaluation not stratified by video $\Rightarrow$ possible leakage
- **Distribution shift between train and test set:**
 - comments in test set are longer (68.6 vs. 58.3 tokens)
 - comments in test set contain higher proportion of Flausch comments (41.3% vs. 29.1%)
 - comments in test set contain more annotated spans per comment (0.65 vs. 0.43).
 - test and train differ in span type distribution.



Thank you!

... and thanks to the
organizers for such
a sweet challenge!

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